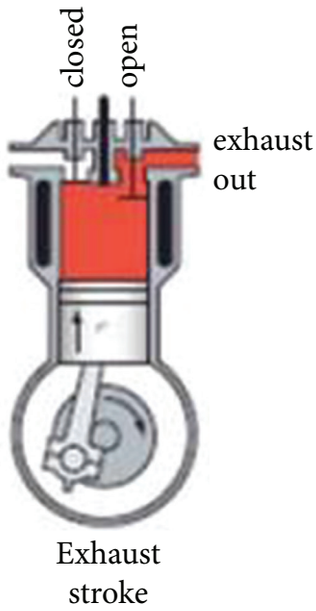


Figure 5.7.1(d) Exhaust Stroke

Scavenging

During the end of exhaust stroke and start of the suction stroke, the inlet valve and exhaust valve will be kept open condition. This will make the fresh charge to push out the burnt out gases from the cylinder and this phenomenon is called as scavenging.

Valve over Lap

The portion of the operating cycle in which both the intake and exhaust valves are kept open is called as valve overlap. It is usually expressed in degrees of crankshaft rotation. Valve overlap is necessary for the efficient flow of gases in and out of the combustion chamber.

Advantages

- ❒ Operating cost is less
- ❒ Lubrication oil consumption is less
- ❒ Can be used in all types of vehicle
- ❒ Thermal Efficiency is high
- ❒ Volumetric Efficiency is high
- ❒ Wear and tear is less

Disadvantages

- ❒ No. of moving parts is more
- ❒ Mechanical Efficiency is low
- ❒ Maintenance cost is more
- ❒ Complicated design
- ❒ More space is required

e) Valve timing diagram of four stroke diesel engine

The valve timing diagram for a four stroke cycle diesel engine is shown in Figure 5.7 (a) below:

In a four stroke diesel engine, an engine cycle to produce power is done on every two revolution of crankshaft. It may be assumed that the valves open and close and injection of fuel occurs at the engine dead centres. However, in actual operation, the valves do not operate at dead centre positions but operate some degree on either side of the dead centres. The opening occurs earlier and the exhaust continues even at later crank angles. The fuel injection is also timed to occur in advance of the completion of compression stroke. The diagram which shows the position of crank of four stroke engine at the beginning and at the end of suction, compression, expansion and exhaust of the engine are called as valve timing diagram. This diagram is known as valve timing diagram.

Inlet valve opening and closing:

In an actual engine, the inlet valve begins to open 10° to 25°before the piston reaches the TDC during the exhaust stroke. This is necessary to ensure that the valve will be fully open when the piston reaches the

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